Jihwan Kwak

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EDUCATION

Seoul National University Sep 2021 – Present
M.S./Ph.D. Integrated Program in Electrical and Computer Engineering (Advisor: Prof. Taesup Moon)
Overall GPA: 3.87 / 4.3

Sungkyunkwan University Mar 2015 – Feb 2021
B.S. in Electrical Engineering
Overall GPA: 4.27 / 4.5

RESEARCH INTERESTS

Core theme: **adaptation**. How models adapt to new data and new context, efficiently.
• Past: Continual learning (class-incremental classification and segmentation) [3, 4, 6]
• Present: Inference efficiency for adaptation (KV cache compression for long-context inference) [1]
• Current: Test-time training (TTT) for long-context models

PUBLICATIONS

(denotes equal contribution)*

[1] **Jihwan Kwak***, Sunghwan Joo, Jung Yoon Hwang, Jaeseok Byun, and Taesup Moon. “KVgrad: Query-Agnostic KV Cache Eviction via Gradient-based Global Importance Scoring.”
- Preprint
- ICML 2026 Workshop on Resource-Adaptive Foundation Model Inference
- KAIA Outstanding Paper Award, KAIA Summer Conference 2026 (CKAIA 2026)
- Algorithm and system-side optimization, including porting to vLLM

[2] Hyunwoong Bae*, **Jihwan Kwak***, and Taesup Moon. “BPEByte: Reusing BPE Vocabulary Priors for Causal and Robust Patching in Hierarchical Byte-level Language Models.”
- Preprint

[3] **Jihwan Kwak**, Sungmin Cha, and Taesup Moon. “Towards Realistic Incremental Scenario in Class Incremental Semantic Segmentation.”
- Conference on Lifelong Learning Agents (CoLLAs), 2024

[4] Sungmin Cha, **Jihwan Kwak**, Dongsub Shim, Hyunwoo Kim, Moontae Lee, and Taesup Moon. “Towards Diverse Evaluation of Class Incremental Learning: A Representation Learning Perspective.”
- Conference on Lifelong Learning Agents (CoLLAs), 2024

[5] Taeeon Park*, **Jihwan Kwak***, Hongjoon Ahn, Jinwoong Lee, Sangho Yu, Changhwan Shin, and Taesup Moon. “GAN-Based Framework for Unified Estimation of Process-Induced Random Variation in FinFET.”
- IEEE Access, 2022

[6] Hongjoon Ahn*, **Jihwan Kwak***, Subin Lim, Hyeonsu Bang, Hyojun Kim, and Taesup Moon. “SS-IL: Separated Softmax for Incremental Learning.”
- International Conference on Computer Vision (ICCV), 2021

[7] Jinwoong Lee, Taeeon Park, Hongjoon Ahn, **Jihwan Kwak**, Taesup Moon, and Changhwan Shin. “Prediction Model for Random Variation in FinFET Induced by Line-Edge-Roughness (LER).”
- Electronics, 2020

PATENTS

[P1] Seokhwan Yoon, **Jihwan Kwak**, Hyunjin Park, Taesup Moon, and Sunghwan Joo. “Electronic Device for Compressing KV Cache of Transformer Model and Operating Method Thereof.”
- Korean Patent Application No. 10-2026-0210274, filed Oct 2026 (pending)
- Joint application of Samsung Electronics and Seoul National University, based on KVgrad [1]

WORK EXPERIENCE

Research Engineer, NAVER, Vision-Language Team Oct 2025 – Mar 2026
- Throughput (TPS) acceleration for multimodal pretraining (Megatron-LM / Megatron-Bridge, NeMo 2)

Visiting Student Researcher, Stanford University (Prof. Tsachy Weissman) Jan 2025 – Feb 2025
- Work on BPEByte [2]

Undergraduate Research Intern, M.IN.D Lab, Seoul National University Jan 2020 – Aug 2021

AWARDS AND HONORS

• KAIA Outstanding Paper Award (CKAIA 2026) 2026
• Best TA Award (C++ Programming), Seoul National University 2022
• Dean’s List and Scholarship, Sungkyunkwan University 2018, 2019, 2020
• First Mover Award and Scholarship, Sungkyunkwan University 2019, 2020
• SKKU President’s List 2019

EXTRACURRICULAR ACTIVITIES

• KATUSA, 19th Expeditionary Sustainment Command, U.S. Army Oct 2016 – Jul 2018
• Supervisor / Interpreter, PyeongChang 2018 Olympic & Paralympic Winter Games Jan 2018 – Mar 2018
• Vice President, Academic Exchange, Young Engineers Honor Society (YEHS) Jan 2020 – Dec 2020

TECHNICAL SKILLS

Languages: Python, C++, Verilog/VHDL
Machine Learning: PyTorch, vLLM, Megatron-LM, NeMo, scikit-learn, NumPy, pandas, Matplotlib